

A STEPWISE MULTIVARIATE t-DISTRIBUTION*

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SUMMARY

A multivariate t-distribution is defined and called a stepwise multivariate t-distribution because it finds application in a form of stepwise regression analysis, namely one in which the variables are included (excluded) in a fixed order, a new regression equation being calculated after each inclusion (exclusion). The non-central distribution also applies to the procedure of testing the slopes of two simple linear regression lines for equality, subsequently fitting parallel lines and testing the intercepts on the y-axis for equality.

1. Introduction

By $t(f)$, $t'(f, \delta)$ and $t''(f, \delta, \lambda)$ are denoted respectively the Student-t distribution, the noncentral t-distribution with noncentrality δ (tabulated in Resnikoff and Lieberman (1957)) and the doubly noncentral t-distribution with non centralities δ and λ (cf. e.g. Krishnan (1967)), each of these distributions having f degrees of freedom (d.f.). δ is of the form Ez/σ where $z \sim N(Ez, \sigma^2)$ and λ is of the form $\lambda = \sum_j (Ez_j/\sigma)^2$ where the z_j are independently $\sim N(Ez_j, \sigma^2)$. (Some authors

write $\lambda = \frac{1}{2} \sum_j (Ez_j/\sigma)^2$.) The chi-square distribution with f d.f. and the noncentral chi-square distribution with f d.f. and noncentrality λ (discussed in Graybill (1961), p. 74 and elsewhere) are denoted by $\chi^2(f)$ and $\chi'^2(f)$ respectively.

Let $S_1^2, z_1, \dots, z_k$ be independently distributed with

$$S_1^2/\sigma^2 \sim \chi^2(f) \text{ and } z_j \sim N(\mu_j, \sigma^2) \text{ for } j = 1, \dots, k$$

and let

$$S_i^2 \equiv S_{i-1}^2 + z_{i-1}^2, i > 1.$$

Then

$$S_i^2 \sim \chi^2(f+i-1) \text{ if } \mu_1 = \dots = \mu_i = 0,$$

while in general

$$S_i^2 \sim \chi'^2(f+i-1, \sum_{j=1}^{i-1} \mu_j^2/\sigma^2) \text{ for } i > 1.$$

Now let

$$t_i = z_i(f+i-1)^{1/2}/S_i.$$

Then

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$$t_i \sim t(f+i-1) \text{ if } \mu_1 = \dots = \mu_i = 0$$

while in general

$$t_i \sim t''(f+i-1, \mu_i/\sigma, \sum_{j=1}^{i-1} \mu_j^2/\sigma^2) \text{ for } i > 1.$$

The joint distribution of $t_1, \dots, t_k$ will be called a stepwise multivariate t-distribution with f basic degrees of freedom (b.d.f.). This distribution is studied in section 2, and $t_1, \dots, t_k$ are shown to be independent in the central case ($\mu_1 = \dots = \mu_k = 0$). An application in stepwise regression analysis, when the order of inclusion (or exclusion) is fixed, is pointed out in section 3, the noncentral bivariate distribution is considered in greater detail in section 4, while power charts are presented in section 5 for purposes of illustration. In section 6 it is shown that the joint distribution of two test statistics commonly used to compare two simple linear regression lines (cf. Hald (1952) pp. 571-574) is a stepwise bivariate t-distribution.

Stepwise procedures in general are discussed by Roy (1958) and Dempster (1963).

2. A Stepwise multivariate t-distribution

Let $z_1, \dots, z_k$ and W be independent random variables with

$$z_i \sim N(\mu_i, \sigma^2), i = 1, \dots, k \text{ and } W \sim \sigma^2 \chi^2(f).$$

If $\mu_1 = \dots = \mu_k = 0$, the joint probability density function (p.d.f.) of

$$z_1/\sigma, \dots, z_k/\sigma, W/\sigma^2 \text{ is}$$

$$p(z_1/\sigma, \dots, z_k/\sigma, W/\sigma^2) = 2^{-1/2(f+k)} \Pi^{-1/2 k} \Gamma(1/2 f)^{-1} W^{1/2 f - 1} e^{-1/2(z_1^2 + \dots + z_k^2 + W)}.$$

Transforming to

$$t_1 = z_1 f^{1/2} W^{-1/2}, t_2 = z_2 (f+1)^{1/2} (W+z_1)^{-1/2}, \dots, t_k = z_k (f+k-1)^{1/2} (W + \sum_{j=1}^{k-1} z_j^2)$$

and integrating out with respect to W from 0 to ∞ , the joint p.d.f. of $t_1, \dots, t_k$ is seen to be

$$(2.1) \quad p(t_1, \dots, t_k) = \prod_{i=0}^{k-1} \Pi^{-1/2} \Gamma(1/2 f + 1/2 i + 1/2) \Gamma(1/2 f + 1/2 i)^{-1} (f+i)^{-1/2} \\ \cdot [1 + t_{i+1}^2 / (f+i)]^{-1/2} (f+i+1).$$

The p.d.f. (2.1) may be written as the product of k univariate Student t p.d.f.'s with $f, f+1, \dots, f+k-1$ d.f. respectively. $t_1, \dots, t_k$ are mutually independent. The multivariate distribution defined by (2.1) will be called the Stepwise k -dimensional Student t -distribution.

If any subset consisting of m of the t_i is chosen and the members of this subset are used simultaneously in t -tests, the individual significance level of each test must be $1 - (1 - \gamma)^{1/m}$ to ensure an overall significance level of γ .

In the noncentral case (some of the $\mu_i \neq 0$) $t_1, \dots, t_k$ are not necessarily independent. The marginal distribution of t_i is

$$t''(f+i-1, \mu_i/\sigma, \sum_{j=1}^{i-1} \mu_j^2/\sigma^2) \text{ for } i > 1$$

and that of t_1 is

$$t'(f, \mu_1/\sigma).$$

The noncentral stepwise bivariate t -distribution is discussed in section 4.

C.G. Troskie (private communication) has drawn my attention to the fact that $t_1, \dots, t_k$ are simply related to special cases of generalized multivariate beta variables. See e.g. Olkin and Rubin (1964).

3. Application in stepwise regression analysis

Consider a form of stepwise regression analysis where the concomitant variables are included in a fixed order. At the i -th step the i -th concomitant variable is included in the regression equation and its regression coefficient is tested. If this coefficient is significantly different from zero, a new regression line is calculated with the next variable included and the process is repeated. The discussion is also valid for a backwards version of this procedure, in which all variables are included initially and deleted one by one. Such tests are used, for example, in a procedure described by Williams (1959), pp. 40-42, to find the degree of a polynomial that fits a set of data. See also Draper and Smith (1966), p. 71.

To illustrate the idea, consider the regression of y on two concomitant variables x_1 and x_2 . The model is assumed to be

$$y_j \sim N\left\{a + \sum_{i=1}^2 \beta_i(x_{ij} - \bar{x}_i), \sigma^2\right\}, j = 1, \dots, n;$$

$$\text{cov}(y_j, y_m) = 0 \text{ if } j \neq m,$$

where

$$\bar{x}_i = \sum_{j=1}^n x_{ij}/n.$$

Define

$$d_{km} = \sum_{j=1}^n (x_{kj} - \bar{x}_k)(x_{mj} - \bar{x}_m) \text{ for } k, m = 1, 2$$

and

$$Y_k = \sum_{j=1}^n y_j(x_{kj} - \bar{x}_k), k = 1, 2.$$

A stepwise build-up of the regression equation is as follows:

Step 0: The constant term is

$$a = \hat{a} = \bar{y} = \sum_{j=1}^n y_j/n$$

while $\text{Var } a = \sigma^2/n$ and the residual sum of squares (RSSQ) is $S_0^2 = \sum_{j=1}^n (y_j - a)^2$.

Step 1: The regression coefficient of x_1 if x_2 is not included, is

$$b_1 = \hat{\beta}_1 = Y_1/d_{11}$$

while $\text{Var } b_1 = \sigma^2/d_{11}$, $\text{cov}(a, b_1) = 0$ and the RSSQ is

$$S_1^2 = \sum_{j=1}^n \{y_j - a - b_1(x_{1j} - \bar{x}_1)\}^2 = S_0^2 - b_1^2 d_{11}.$$

Step 2: The regression equation of y on x_1 and x_2 is

$$y = a + b'_1(x_1 - \bar{x}_1) + b'_2(x_2 - \bar{x}_2),$$

where $b'_1 = (d_{22}Y_1 - d_{12}Y_2)/(d_{11}d_{22} - d_{12}^2)$, $b'_2 = (d_{11}Y_2 - d_{12}Y_1)/(d_{11}d_{22} - d_{12}^2)$
Now $\text{Var } b'_2 = \sigma^2 d_{11}/(d_{11}d_{22} - d_{12}^2)$ and $\text{cov}(a, b'_i) = 0$ for $i = 1, 2$. Also, b_1 as calculated in step 1, may be written

$$b_1 = b'_1 + d_{12}b'_2/d_{11}$$

and it follows that $\text{cov}(b_1, b'_2) = 0$. The RSSQ is

$$\begin{aligned} S_2^2 &= \sum_{j=1}^n \left\{ (y_j - a - \sum_{i=1}^2 b_i(x_{ij} - \bar{x}_i)) \right\}^2 \\ &= S_1^2 - b_2'^2 (d_{11}d_{22} - d_{12}^2)/d_{11}. \end{aligned}$$

A stepwise trivariate t -distribution may be defined as follows:

$$t_1 = b'_2 \{ (d_{11}d_{22} - d_{12}^2)/d_{11} \}^{1/2} (n-3)^{-1/2} S_2^{-1},$$

the t -variable used for testing $H_0 : \beta_2 = 0$;

$$t_2 = b_1 d_{11}^{1/2} (n-2)^{-1/2} S_1^{-1},$$

the t -variable used for testing $H_0 : \beta_1 = 0$ under the assumption that $\beta_2 = 0$; and

$$t_3 = a n^{1/2} (n-1)^{-1/2} S_0^{-1},$$

the t -variable used for testing $H_0 : E y = 0$ under the assumption that $\beta_1 = \beta_2 = 0$.

The joint distribution of t_1, t_2, t_3 is seen to be a stepwise trivariate distribution discussed in section 2. If not all of α, β_1 and β_2 are equal to zero, then t_1 has a noncentral t -distribution while t_2 and t_3 have doubly noncentral t -distributions.

This process may be continued for more concomitant variables but the algebra becomes tedious. D.J. de Waal (private communication) suggested transforming the concomitant variables into orthogonal variables to simplify the algebra, and M.W. Browne in his comment on the paper at the annual conference suggested using Bartlett's decomposition of the covariance matrix.

The purpose in this paper is not to propagate the use of this method for determining e.g. the degree of a polynomial that fits a given set of data, but rather to throw some light on a method often suggested in textbooks. A more general study of the problem is reported in Anderson (1962), which I was referred to by C.G. Troskie.

4. A Noncentral stepwise bivariate t -distribution

Let z_1, z_2 and W be independent with $W/\sigma^2 \sim \chi^2(f)$ and $z_i \sim N(\mu_i, \sigma^2)$, $i = 1, 2$ and define

$$(4.1) \quad t_1 = z_1 f^{1/2} W^{-1/2}, \quad t_2 = z_2 (f+1)^{1/2} (W + z_1^2)^{-1/2}.$$

If $\mu_1 = \mu_2 = 0$ the joint p.d.f. of t_1 and t_2 is the product of two univariate Student t -p.d.f.'s with f and $f+1$ d.f. respectively. In order to ensure a joint significance level γ , each t_i must be tested at the level $1 - (1-\gamma)^{1/2}$.

If $\mu_1 \neq 0, \mu_2 \neq 0$ the joint p.d.f. of $z_1/\sigma, z_2/\sigma$ and W/σ^2 may be written

$$p(z_1/\sigma, z_2/\sigma, W/\sigma^2) = 2^{-1/2f-1} \Pi^{-1} \Gamma(1/2 f)^{-1} e^{-1/2(\delta_1^2 + \delta_2^2)} W^{1/2 f-1} \\ e^{-1/2(z_1^2 + z_2^2 + W)} \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} z_1^i \delta_1^j (j!)^{-1} z_2^i \delta_2^i (i!)^{-1},$$

where $\delta_i = \mu_i/\sigma, i = 1, 2$.

Applying transformation (4.1) and integrating out for W from 0 to ∞ , the joint p.d.f. of t_1 and t_2 is seen to be

$$(4.2) \quad p(t_1, t_2) = \Pi^{-1} \Gamma(1/2 f)^{-1} e^{-1/2(\delta_1^2 + \delta_2^2)} \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} (2^{1/2} \delta_1)^j f^{-1/2 j - 1/2} (j!)^{-1} \\ (2^{1/2} \delta_2)^i (f+1)^{-1/2 i - 1/2} (i!)^{-1} \Gamma(1/2 f + 1/2 i + 1/2 j + 1) \\ t_1^j \{1 + t_1^2/f\}^{-1/2(f+j+1)} t_2^i \{1 + t_2^2/(f+1)\}^{-1/2(f+i+j+2)}.$$

Therefore

$$(4.3) \quad P_2(t_1^0, t_2^0) = \Pr\{|t_1| < t_1^0, |t_2| < t_2^0\} \\ = \sum_{j=0}^{\infty} \text{Psn}(j, 1/2 \delta_1^2) I_{T_1^0}(j+1/2, 1/2 f) \\ \cdot \sum_{i=0}^{\infty} \text{Psn}(i, 1/2 \delta_2^2) I_{T_2^0}(i+1/2, 1/2 f + j + 1/2)$$

where $I_X(m, n)$ is the incomplete beta-function ratio, $\text{Psn}(m, \lambda) = e^{-\lambda} \lambda^m / m!$ and $T_i^0 = t_i^{02} / (f + i - 1 + t_i^{02}), i = 1, 2$.

The marginal two-sided probabilities of t_1 and t_2 are found by substituting $T_2^0 = 1$ and $T_1^0 = 1$ respectively in (4.3). Thus

$$(4.4) \quad P_1(t_1^0) = \Pr\{|t_1| < t_1^0\} = \sum_{j=0}^{\infty} \text{Psn}(j, 1/2 \delta_1^2) I_{T_1^0}(j+1/2, 1/2 f),$$

which corresponds to the central probability of the noncentral t-distribution, and

$$(4.5) \quad P_2(t_2^0) = \Pr\{|t_2| < t_2^0\} = \sum_{j=0}^{\infty} \text{Psn}(j, 1/2 \delta_1^2) \\ \cdot \sum_{i=0}^{\infty} \text{Psn}(i, 1/2 \delta_2^2) I_{T_2^0}(i+1/2, 1/2 f + j + 1/2),$$

which is the central probability of the doubly noncentral t-distribution (cf. Weibull (1953) where the corresponding result is given for the doubly noncentral F-distribution.)

Since $P_{12}(t_1^0, t_2^0) \neq P_1(t_1^0)P_2(t_2^0)$, t_1 and t_2 are seen to be dependent.

If $\mu_1 = 0, \mu_2 \neq 0$, t_1 and t_2 are seen to be independent with $t_1 \sim t(f)$ and $t_2 \sim t'(f+1, \delta_2)$. If $\mu_1 \neq 0, \mu_2 = 0$, t_1 and t_2 are dependent, with $t_1 \sim t'(f, \delta_1)$ while t_2 has the noncentral distribution discussed by Marakathavalli (1954).

5. Power charts

The following four probability integrals are of interest:

$$I_1 = 1 - P_{12}(t_1^0, t_2^0) = \Pr\{|t_1| > t_1^0 \text{ or } |t_2| > t_2^0\};$$

$$I_2 = 1 - P_1(t_1^0) = \Pr\{|t_1| > t_1^0\};$$

$$I_3 = 1 - P_2(t_2^0) = \Pr\{|t_2| > t_2^0\}; \text{ and}$$

$$I_4 = I_1 - I_2 = \Pr\{|t_1| < t_1^0 \text{ and } |t_2| > t_2^0\}.$$

The last integral is of interest because in some procedures t_2 is tested only if t_1 is not significant.

P_{12} , P_1 and P_2 , as defined in (4.3)-(4.5), were calculated by summing in each direction (i and j) until the series are found to be decreasing and the terms are smaller than 10^{-6} . The numerical integrals were confirmed in a few cases by means of a Monte Carlo study.

I_1 , the power of the whole test procedure, i.e. the joint power of t_1 and t_2 , is depicted in Fig. 1 for 10 b.d.f., noncentralities $\lambda_i = \mu_i^2/\sigma^2 = \delta_i^2$, $i = 1, 2$ and significance level $\gamma = 0.05$. It is interesting to note that for large values of λ_2 , the power decreases initially with increasing values of λ_1 , which shows that the increase in power of t_1 is initially not adequate to counteract the decrease in power of t_2 as λ_1 increases.

I_2 , the power of t_1 , is depicted in Fig. 2 for overall significance level 0.05 and a collection of d.f. I_2 is the power of the univariate two-sided t-test with significance level $1 - 0.95^{1/2} \doteq 0.02532$.

I_3 and I_4 are depicted in Fig. 3 for 10 b.d.f. which gives 11 d.f. for t_2 . This shows the decrease in power of a t-test if a noncentrality is present in the denominator. I_3 may be looked upon as the power of the univariate two-sided t-test with significance level approximately 0.02532 when noncentralities occur in the numerator as well as the denominator.

6. Application in univariate analysis of covariance

Two simple linear regression lines are often compared in the following way (cf. Hald (1952) pp. 571-574):

Given are two sets of observations (x_{ij}, y_{ij}) , $j = 1, \dots, n_i$; $i = 1, 2$ where the x_{ij} are chosen values of the nonrandom variable x and the y_{ij} are independent observations on the normally distributed random variable y such that

$$E(y_{ij}|x_{ij}) = \eta_i(x_{ij}) = \alpha_i + \beta_i(x_{ij} - \bar{x}_i),$$

$$\text{Var}(y_{ij}|x_{ij}) = \sigma^2, \text{ where } \bar{x}_i = \sum_{j=1}^{n_i} x_{ij}/n_i.$$

Write

$$d_i = \sum_j (x_{ij} - \bar{x}_i)^2; \quad \bar{y}_i = \sum_j y_{ij}/n_i.$$

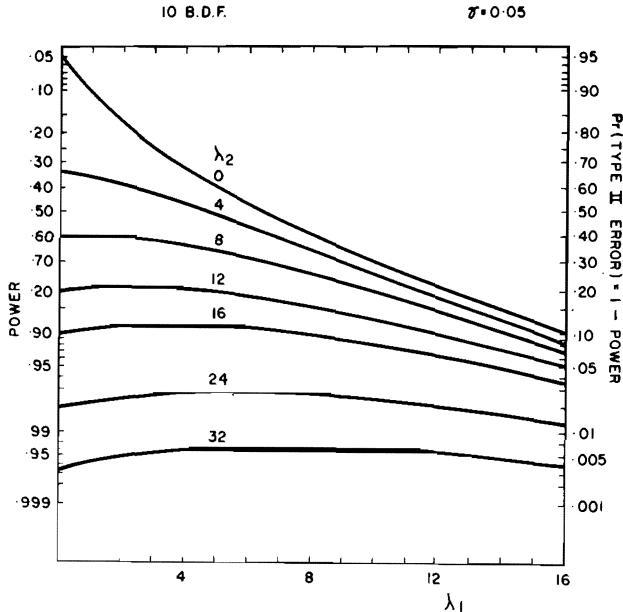


FIG. 1 Power of the two-sided bivariate stepwise t -test with 10 basic degrees of freedom, noncentralities $\lambda_i = \mu_i^2/\sigma^2$, $i=1,2$ and significance level $\gamma = 0.05$.

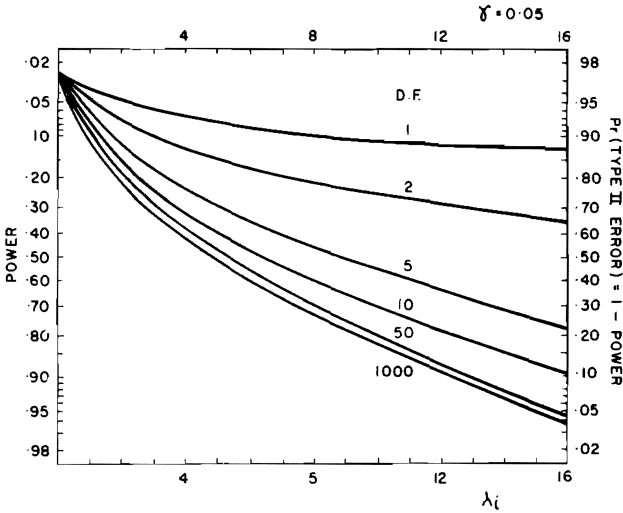


FIG. 2 Power of the two-sided t -test with noncentrality $\lambda_i = \mu_i^2/\sigma^2$, $i=1$ or 2 , and significance level $1 - \sqrt{1 - \gamma} = 1 - \sqrt{0.95}$

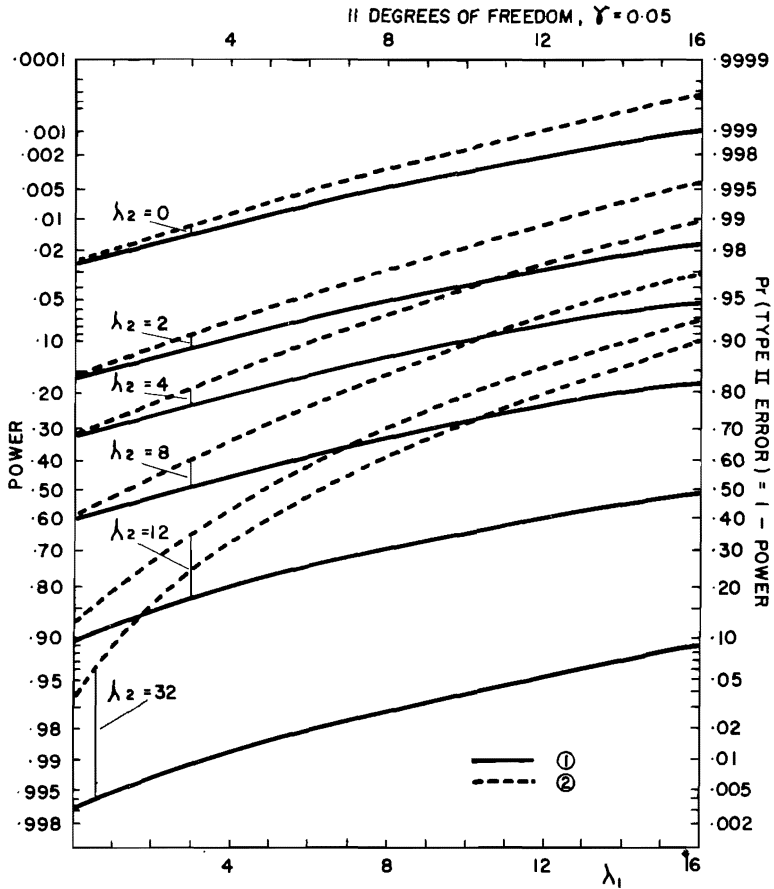


FIG. 3 Power of the second member of the two-sided bivariate t -test with 10 basic degrees of freedom, noncentralities $\lambda_i = \mu_i^2 / \sigma^2$, $i = 1, 2$ and overall significance level $\gamma = 0.05$ if ① tested irrespective of the outcome of the first test; ② tested only if the first part of the null hypothesis is not rejected.

The least squares estimates of a_i and β_i (cf. Hald (1952) pp. 528-541) are $a_i = \hat{a}_i = \bar{y}_i$; $b_i = \hat{\beta}_i = \sum_j y_{ij}(x_{ij} - \bar{x}_i)/d_i$, while σ^2 is estimated by $s^2 = S^2/(n_1 + n_2 - 4)$ where

$$S^2 = \sum_i \sum_j \{y_{ij} - a_i - b_i(x_{ij} - \bar{x}_i)\}^2.$$

The null hypothesis $H_0(1) : \beta_1 = \beta_2$ is rejected if $|t_1| > t_1^0$ where $t_1 = (b_1 - b_2)s^{-1}(1/d_1 + 1/d_2)^{-1/2}$ and t_1^0 is a critical value of the Student t-distribution with $n_1 + n_2 - 4$ d.f.

If $H_0(1)$ is not rejected, the lines are assumed to be parallel and the parallel lines are estimated by calculating a common slope $b = \hat{\beta} = (b_1 d_1 + b_2 d_2)/(d_1 + d_2)$, while σ^2 is estimated by $s_p^2 = S_p^2/(n_1 + n_2 - 3)$ where

$$\begin{aligned} S_p^2 &= \sum_i \sum_j \{y_{ij} - a_i - b(x_{ij} - \bar{x}_i)\}^2 \\ &= S^2 + (b_1 - b_2)^2/(1/d_1 + 1/d_2). \end{aligned}$$

The null hypothesis of equal intercepts on the y-axis, i.e. $H_0(2) : \eta_1(0) = \eta_2(0)$ is rejected if $|t_2| > t_2^0$ where

$$t_2 = (a_1 - b\bar{x}_1 - a_2 + b\bar{x}_2)s_p^{-1} \{1/n_1 + 1/n_2 + (\bar{x}_1 - \bar{x}_2)^2/(d_1 + d_2)\}^{-1/2}$$

and t_2^0 is a critical value of the Student t-distribution with $n_1 + n_2 - 3$ d.f. The first factor of t_2 may be written

$$(a_1 - b\bar{x}_1) - (a_2 - b\bar{x}_2) = a_1 + b_1(x_0 - \bar{x}_1) - a_2 - b_2(x_0 - \bar{x}_2)$$

where $x_0 = (d_2 \bar{x}_1 + d_1 \bar{x}_2)/(d_1 + d_2)$, and it is easily seen that this quantity is independent of $b_1 - b_2$, so that t_1 and t_2 have the stepwise bivariate t-distribution with $n_1 + n_2 - 4$ b.d.f.

A stepwise multivariate t-distribution would occur when two multivariate regression lines are compared : at the k-th step H_0 : "The regression coefficients with respect to concomitant variable number k are equal" is tested. If H_0 is not rejected, new regression lines are calculated with a common regression coefficient with respect to concomitant variable number k.

REFERENCES

- ANDERSON, T.W. (1962). The choice of the degree of a polynomial regression as a multiple decision problem. *Ann. Math. Statist.* **33**, 255-265.
 DEMPSTER, A.P. (1963). Multivariate theory for general stepwise methods. *Ann. Math. Statist.* **34**, 873-883.
 DRAPER, N.R. and SMITH, H. (1966). *Applied regression analysis*. John Wiley, New York.
 GRAYBILL, Franklin A. (1961). *An introduction to linear statistical models*, Vol. 1. McGraw-Hill, New York.
 HALD, A. (1952). *Statistical theory with engineering applications*. John Wiley, New York.

- KRISHNAN, Marakatha (1967). The moments of a doubly noncentral t-distribution. *J. Amer. Statist. Assoc.* 62, 278-287.
- MARAKATHAVALLI, N. (1954). The distribution of t_1 and its applications. *J. Madras Univ. B* 24, 251-272.
- OLKIN, Ingram and RUBIN, Herman (1964). Multivariate beta distributions and independence properties of the Wishart distribution. *Ann. Math. Statist.* 35, 261-269.
- RESNIKOFF, George J. and LIEBERMAN, Gerald J. (1957). *Tables of the non-central t-distribution*. Stanford Univ. Press, Stanford.
- ROY, J. (1958). Step-down procedure in multivariate analysis. *Ann. Math. Statist.* 29, 1177-1187.
- WEIBULL, Martin (1953). The distribution of t- and F-statistics and of correlation and regression coefficients in stratified samples from normal populations with different means. *Skand. Akt. Tidskr.* 36 Suppt. 9-106.
- WILLIAMS, E.J. (1959). *Regression analysis*. John Wiley, New York.

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